patch wednesday

git + fixing bugs — the one-page cheat sheet

the loop (do this per bug)

1. open the bug's file in `challenges/` — it explains the bug + fix
2. fix the one function it points at
3. `git add .`
4. `git commit -m "fix the login query"`
5. `git push`
6. github **actions** tab → one less red = it worked

the five git commands

```
$ git clone <link>   # once, to get your copy
$ git status         # what changed
$ git diff           # read it before you commit
$ git add .
$ git commit -m "..."
$ git push           # sends it; tests run
```

broke it?
`git restore <file>` — undo your edits, back to the last commit.

red cross?
that just means a bug is still there. open actions → pytest to see which test fails and why.

reading a failure

```
FAILED tests/test_notes.py
  alice should not be able
  to read bob's note
```

the message tells you what's still wrong. the file's comment tells you the fix.

the bugs

login.py — sql injection. don't build the query from text; use `?` placeholders.

notes.py — idor. check the note belongs to you before showing it.

files.py — path traversal. keep the download inside its folder.

tools.py (bonus) — command injection. no shell; pass args as a list.

redirect_after_login.py (bonus) — open redirect. allow only local paths.

done early?
swap repo links with a neighbour and try to break their fix (open an issue if you do). then hunt: there are at least 3 more bugs the tests don't check.

golden rule
fix the **whole class** of bug, not the one input. blocklists don't pass.